

● RAG Learning

Topic 5: Health, Disease and the Development of Medicines

Edexcel GCSE (9–1) Biology (1BIO) — Paper 2

5.1 What is Health?

The **World Health Organisation (WHO)** defines health as *"a state of complete physical, mental and social wellbeing and not merely the absence of disease or infirmity."*

Health is more than simply being free of illness. It has three components, all of which must be reasonably good for a person to be truly healthy:

- **Physical wellbeing** — bodies functioning normally; absence of injury or disease.
- **Mental wellbeing** — ability to think clearly, cope with stress, handle emotions.
- **Social wellbeing** — positive relationships, sense of belonging, ability to interact with others.

Disease

A **disease** is a condition that impairs the normal functioning of the body. Diseases are split into two broad categories:

Category	Cause	Spread?	Examples
Communicable (infectious)	Caused by a pathogen — a microorganism that invades the body	Yes — passes from one organism to another (direct contact, water, air,	Cholera, TB, malaria, HIV, flu, measles, chalara ash dieback.

	(bacteria, viruses, fungi, protists).	vectors, body fluids).	
Non-communicable	Caused by genetics, lifestyle, age or environment; not caused by a pathogen.	No — cannot be passed between individuals.	Cardiovascular disease, most cancers, type 2 diabetes, asthma, obesity-related conditions.

One disease increasing the risk of another

Diseases do not occur in isolation. One illness can make a person more likely to suffer from another:

- A **compromised immune system** (e.g. HIV infection, which destroys lymphocytes) leaves the body vulnerable to infections that a healthy immune system would clear easily — TB, pneumonia, certain cancers.
- Some **viruses trigger cancers**: HPV (cervical cancer), hepatitis B/C (liver cancer), Epstein–Barr virus (certain lymphomas).
- Long-term **immune responses to pathogens** can trigger allergies or autoimmune problems.
- Severe physical illness affects **mental health** (depression/anxiety), which in turn can worsen recovery and impair immune function.

Exam Tip: A common 2-mark question asks you to give the WHO definition of health and to give one example of how one disease can increase the risk of another. Use a named pathogen (e.g. *HIV weakens the immune system so a person is more likely to catch TB*).

5.2 Pathogens

A **pathogen** is a microorganism that causes a communicable disease. Pathogens fall into four groups: **bacteria, viruses, fungi and protists**.

Pathogen type	Cell?	Size	How it causes disease	Example disease
Bacteria	Prokaryotic cell	~1–5 μm	Reproduce rapidly in the body; many release toxins that damage cells and tissues.	Cholera, TB, stomach ulcers (<i>H. pylori</i>)
Viruses	Not a cell — just genetic material in a protein coat	~20–300 nm (tiny)	Invade host cells, hijack their machinery to make new viruses; host cell bursts and dies.	HIV, influenza, Ebola, measles
Fungi	Eukaryotic (often multicellular hyphae)	A few μm to visible	Grow through tissue (hyphae), releasing spores that spread and extracellular enzymes that digest tissue.	Chalara ash dieback, athlete's foot
Protists	Eukaryotic single-celled	~10–100 μm	Live inside host cells or body fluids, often transferred by a vector (e.g. mosquito). Damage tissues directly.	Malaria (<i>Plasmodium</i>)

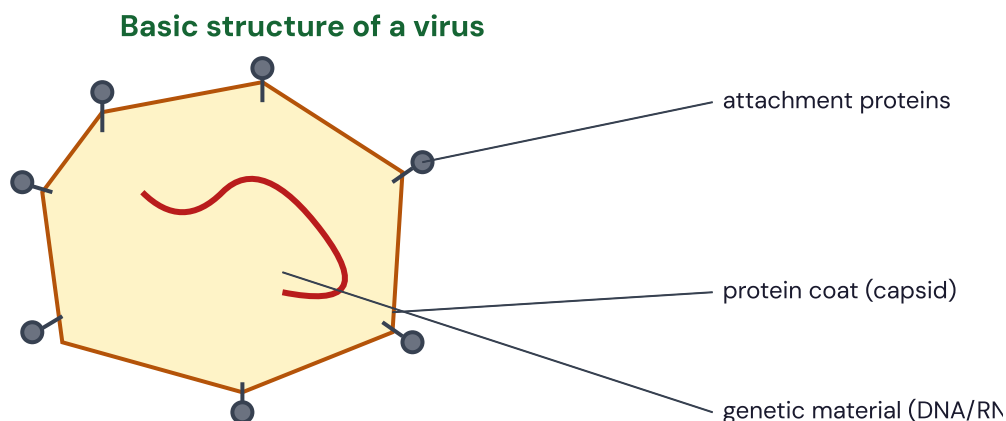
5.3 Viruses

Viruses are **not alive** in the usual sense — they are not made of cells, do not respire, do not grow and cannot reproduce on their own. They can only replicate inside a living host cell.

Structure of a virus

- **Genetic material** — either DNA or RNA, carrying instructions to make new viruses.
- **Protein coat (capsid)** — surrounds and protects the genetic material.

- Some viruses have an outer **envelope** of lipid (taken from the host cell membrane) studded with proteins used to attach to host cells.



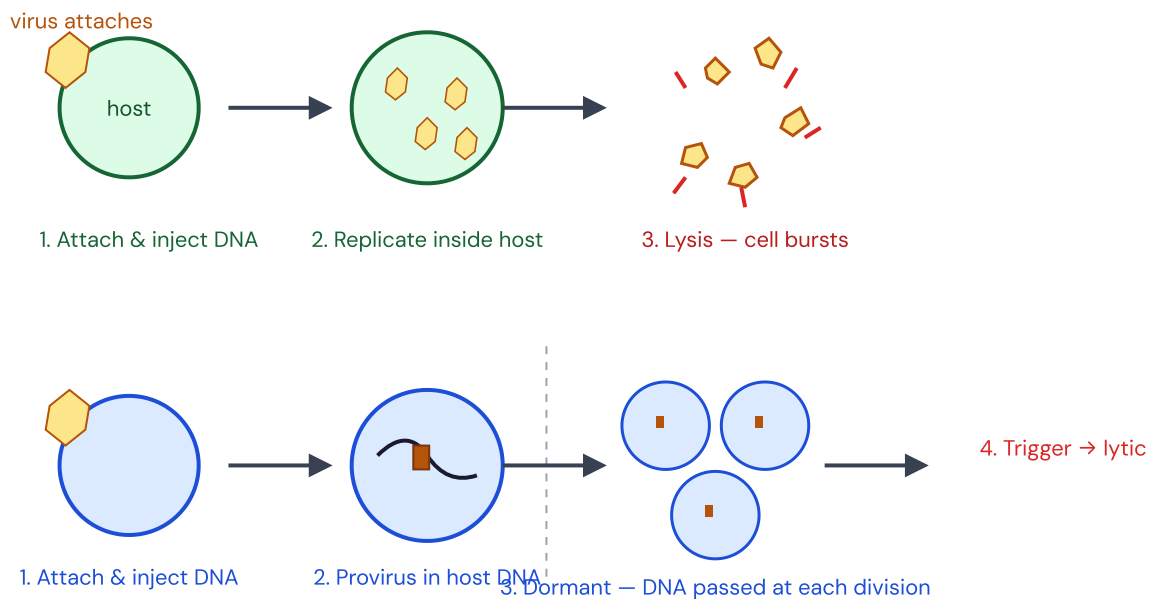
The lytic cycle vs the lysogenic cycle

Once a virus enters a host cell, it can reproduce in two ways. In the **lytic cycle** the virus immediately makes copies of itself and bursts the host cell open. In the **lysogenic cycle** the viral genetic material is inserted into the host's DNA and lies dormant for many cell divisions, until triggered to enter the lytic cycle later.

Stage	Lytic cycle	Lysogenic cycle
1. Attachment	Virus attaches to receptors on host cell surface.	Same.
2. Entry	Viral genetic material is injected into host cell.	Same.
3. Synthesis	Host cell machinery forced to copy viral DNA/RNA and make viral proteins.	Viral DNA is integrated into host chromosome (a provirus) and replicated each time the host divides.
4. Assembly	New viruses self-assemble from the copied parts.	Dormant — no new viruses made.
5. Release	Host cell lyses (bursts) releasing new viruses; host dies.	Under stress (UV, chemicals, poor host health) the provirus activates and the cycle

switches to lytic.

Lytic cycle (left) vs Lysogenic cycle (right)



Exam Tip: Lytic = lysis (cell bursts, virus reproduces immediately). Lysogenic = integrated into host DNA, stays hidden. Herpes and HIV both have lysogenic stages — this is why these infections are very hard to get rid of.

5.4 Communicable Diseases — The Edexcel Case Studies

You must know the following named diseases: the pathogen type, how it spreads, its main symptoms, and how it can be prevented or treated.

Cholera

Case study: Pathogen: *Vibrio cholerae* (bacterium).

Transmission: consumption of food or water contaminated by faeces containing the bacterium — common where there is poor sanitation.

Symptoms: severe watery **diarrhoea** (the bacterium releases a toxin that makes gut cells pump water out), dehydration — can be fatal in hours.

Prevention: clean drinking water, sewage treatment, hand-washing, oral rehydration therapy (ORT) for treatment.

Tuberculosis (TB)

Case study: Pathogen: *Mycobacterium tuberculosis* (bacterium).

Transmission: **airborne droplets** when an infected person coughs or sneezes.

Symptoms: persistent cough (sometimes with blood), chest pain, fever, weight loss ("consumption"); primarily damages the lungs but can spread.

Prevention/treatment: BCG vaccine in high-risk areas, good ventilation, avoiding crowded indoor spaces, long course of antibiotics (6 months). Risk groups include people with HIV.

Chalara Ash Dieback

Case study: Pathogen: *Hymenoscyphus fraxineus* (fungus).

Transmission: **spores carried in the wind**; can also spread via movement of infected saplings between countries.

Symptoms (in ash trees): leaf loss, dark brown/black lesions on bark, crown dieback, eventual tree death.

Prevention: restricting movement of ash trees, removing and burning infected trees, researching resistant varieties.

Malaria

Case study: Pathogen: *Plasmodium* (protist).

Transmission: bite of an infected female *Anopheles* mosquito — the mosquito is the **vector**.

Symptoms: recurring fever, chills, sweating, anaemia; liver damage; can be fatal.

Prevention: insecticide-treated bed nets, indoor insecticide spraying, removing standing water where mosquitoes breed, antimalarial drugs for travellers.

HIV / AIDS

Case study: Pathogen: Human Immunodeficiency Virus (HIV).

Transmission: exchange of body fluids — unprotected sex, sharing needles, mother to baby (in pregnancy, birth or breastfeeding), infected blood transfusions.

Symptoms: initial flu-like illness; virus attacks T-lymphocytes (white blood cells); after years, the immune system is so weakened that the person develops **AIDS** and cannot fight off other infections (e.g. TB, pneumonia).

Prevention/treatment: condoms, clean needles, screening of blood donations, antiretroviral (ART) drugs keep viral load low but cannot cure.

Stomach Ulcers — *Helicobacter pylori*

Case study: Pathogen: *Helicobacter pylori* (bacterium).

Transmission: oral–oral or faecal–oral spread (saliva, contaminated food/water).

Symptoms: stomach pain, nausea, indigestion; bacterium burrows into the stomach lining, producing ammonia which damages cells and leads to an ulcer (also a cancer risk factor).

Treatment: combination antibiotics plus a proton–pump inhibitor to reduce stomach acid.

Ebola

Case study: Pathogen: Ebola virus.

Transmission: direct contact with the **body fluids** (blood, vomit, saliva) of an infected person or animal; can remain infectious in a body after death.

Symptoms: severe haemorrhagic fever — fever, severe internal and external bleeding, organ failure; high mortality (~50%).

Prevention: isolation of cases, protective equipment for carers, safe burial practices, experimental vaccines.

Summary comparison table

Disease	Pathogen type	Transmission	Main symptoms	Main prevention
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Cholera	Bacterium	Contaminated water	Severe diarrhoea, dehydration	Clean water, sanitation, ORT
TB	Bacterium	Airborne droplets	Cough, weight loss, chest pain	BCG vaccine, ventilation, antibiotics
Ash dieback	Fungus	Wind-borne spores	Leaf loss, dark lesions (ash trees)	Remove infected trees, resistant varieties
Malaria	Protist (<i>Plasmodium</i>)	Mosquito vector	Fever, chills, anaemia	Nets, insecticides, drain standing water
HIV/AIDS	Virus	Body fluids, sex, needles	Immune collapse → AIDS	Condoms, clean needles, ART drugs
Stomach ulcers	Bacterium (<i>H. pylori</i>)	Oral/faecal–oral	Stomach pain, ulceration	Antibiotics + PPIs
Ebola	Virus	Body fluids	Haemorrhagic fever	Isolation, PPE, safe burial

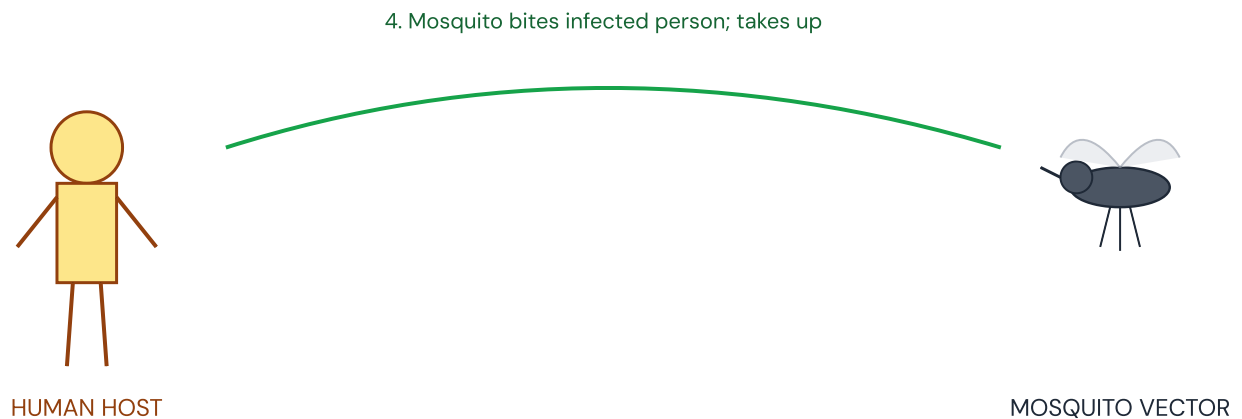
5.5 The Malaria Life Cycle

Plasmodium is a single-celled **protist** with a complex life cycle that requires two hosts: a human and a female *Anopheles* mosquito. The mosquito is the **vector** — an organism that carries the pathogen between hosts but is not itself severely harmed.

1. An infected female *Anopheles* mosquito bites a human to take a blood meal. Her saliva contains *Plasmodium* cells, which enter the human's blood.
2. *Plasmodium* travels to the **liver**, multiplies there for about a week, then returns to the blood.
3. It invades **red blood cells**, multiplies inside them and makes the cells burst — causing the classic fever pattern and anaemia of malaria.

4. Another mosquito bites the infected person, taking in *Plasmodium* cells with the blood.
5. Inside the mosquito's gut, *Plasmodium* reproduces sexually and migrates to the salivary glands, ready to infect the next human.

Malaria life cycle — human and mosquito stages



Plasmodium 1. Infected mosquito bites human; injects *Plasmodium* in saliva 2. Liver: *Plasmodium* multiplies 3. Invades red blood cells → burst 5. Reproduces in gut Moves to salivary glands

Exam Tip: When asked how malaria is prevented, mention preventing the vector not just treating the patient — insecticide-treated nets, draining stagnant water (where larvae grow), indoor spraying, and avoiding mosquito bites with clothing/repellent.

5.6 Plant Defences Against Pathogens

Plants are attacked by bacteria, viruses, fungi and protists too — and they cannot run away or mount an immune response like animals. Instead, plants use a combination of **physical barriers** and **chemical defences**.

Physical defences

- **Waxy cuticle** — a waterproof layer on leaves that stops pathogens landing on the surface from entering.

- **Cellulose cell walls** — a tough physical barrier around every plant cell that most pathogens cannot cross.
- **Bark/dead cells on stems** — a thick outer layer of dead, lignified cells that pathogens cannot easily penetrate; old bark flakes off, taking pathogens with it.
- **Stomatal closure** — stomata (pores for gas exchange) can close when a pathogen is detected to stop it entering.

Chemical defences

- **Antibacterial / antifungal compounds** — many plants make chemicals that kill microbes (e.g. mint contains menthol; witch hazel contains tannins).
- **Poisons that deter herbivores** — *nicotine* (tobacco), *caffeine* (coffee, tea), *pyrethrum* (chrysanthemums) — all evolved to poison insect pests that would otherwise damage the plant and open it to infection.

Identifying plant diseases

Plant pathologists use three main approaches to diagnose a sick plant:

1. **Observation** — visible symptoms such as leaf spots, wilting, stunted growth, discolouration, unusual growths (galls), rotting or "dieback" of branches.
2. **Laboratory testing** — take a tissue sample, look under a microscope for the pathogen, or use DNA tests (PCR) and monoclonal antibody kits to identify the exact species.
3. **Distribution analysis** — map where infected plants occur. A disease spreading through adjacent plants suggests a contact/vector pathogen; clusters downwind suggest spores in air; patchy distribution may indicate soil-borne pathogens.

5.7 Sexually Transmitted Infections (STIs)

A **sexually transmitted infection (STI)** is a communicable disease passed mainly through sexual contact — the exchange of body fluids during intercourse.

HIV

The Human Immunodeficiency Virus was covered in 5.4. It infects and gradually destroys **T-helper lymphocytes**, which normally coordinate the immune response. As T-helper numbers fall, the immune system loses its ability to fight off other infections. The person develops **AIDS** (Acquired Immune Deficiency Syndrome) and is vulnerable to opportunistic infections and cancers.

Chlamydia

Chlamydia trachomatis is actually a bacterium, but it behaves unusually — it can only reproduce inside host cells, like a virus.

- **Transmission:** unprotected vaginal, anal or oral sex with an infected partner; also from mother to baby during birth.
- **Symptoms:** often asymptomatic (especially in women) — which is why it spreads so widely. When symptoms appear they include pain on urinating, unusual discharge, pelvic pain.
- **Consequences:** if untreated can cause pelvic inflammatory disease and infertility.
- **Treatment:** antibiotics (it is a bacterium, so antibiotics work).

Preventing STIs

- **Condoms** — a physical barrier that prevents exchange of body fluids; the single most effective method alongside abstinence.
- **Screening and testing** — routine tests allow early diagnosis and treatment before transmission to partners.
- **Monogamy / reducing partner numbers** — reduces exposure.
- **Abstinence** — the only 100% effective method.
- **Contact tracing** — when someone is diagnosed, past partners are informed so they can be tested.
- **Education** — especially of young people about risks and methods of prevention.

5.8 The Human Body's Non-Specific Defences

Before a pathogen ever reaches the bloodstream, the body has a battery of physical and chemical barriers designed to stop it getting in. These are **non-specific** — they defend against

any pathogen rather than a specific one.

Defence	Location	How it works
Skin	Outer surface of the body	Thick layer of dead, keratinised cells forms a physical barrier; secretes sebum (slightly acidic and contains antimicrobial chemicals); skin bacteria compete with pathogens.
Mucus & cilia	Lining of airways (trachea, bronchi)	Sticky mucus traps pathogens and dust. Ciliated cells beat the mucus up the airways to the throat, where it is swallowed into stomach acid.
Stomach acid (HCl)	Stomach	Strongly acidic (pH ~2) — kills most bacteria and viruses swallowed in food, drink and mucus.
Tears (lysozyme)	Eye surface	Tears contain lysozyme , an enzyme that breaks down bacterial cell walls. Blinking washes pathogens away.
Nasal hairs & mucus	Nose	Hairs filter out large particles; mucus traps pathogens and is expelled by sneezing or swallowed.
Blood clotting	Wounds	Platelets and clotting proteins form a scab, sealing breaks in the skin before pathogens can enter.

Exam Tip: When describing defences, be sure to separate **physical** (skin, mucus, hairs, cilia, scab) from **chemical** (lysozyme in tears, stomach acid, sebum). Examiners like seeing both types listed.

5.9 The Immune System

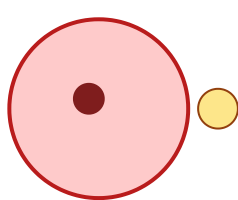
If pathogens get past the outer defences, the **immune system** — largely the **white blood cells** — takes over. There are two main types of white blood cell on the specification: phagocytes and lymphocytes.

Phagocytes — non-specific "eaters"

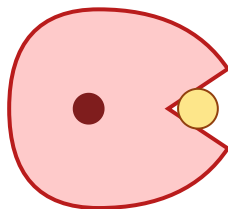
Phagocytosis is the process by which a phagocyte engulfs and digests a pathogen.

1. The phagocyte recognises foreign chemicals on the pathogen's surface (non-specific).
2. It flows round the pathogen and engulfs it, taking it inside a vesicle called a *phagosome*.
3. Lysosomes inside the phagocyte fuse with the vesicle and release enzymes that digest the pathogen.
4. Harmless breakdown products are released or displayed on the surface.

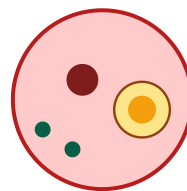
Phagocytosis — four stages



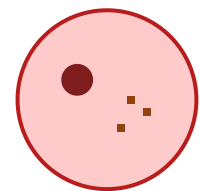
1. Recognition



2. Engulfing



3. Enzymes digest



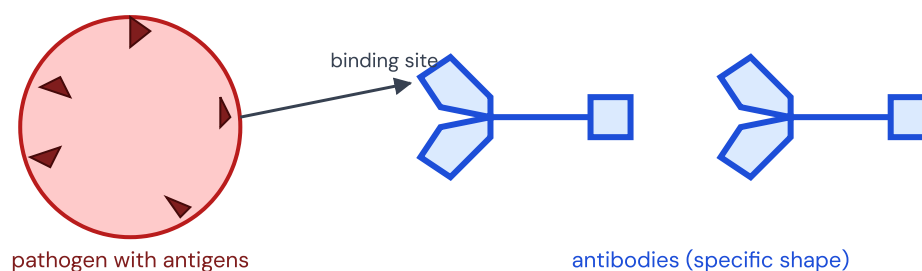
4. Debris released

Lymphocytes — specific "targeters"

Lymphocytes are the **specific** part of the immune system. Every pathogen has unique chemical markers on its surface called **antigens**. Lymphocytes produce **antibodies** — Y-shaped proteins with binding sites that exactly fit the shape of one specific antigen, like a key in a lock.

1. A lymphocyte recognises an antigen on the pathogen.
2. It divides rapidly (*clonal expansion*) to make a clone of cells that all make the same antibody.
3. Antibodies bind to antigens, causing pathogens to clump together (*agglutination*), or marking them for phagocytes, or neutralising their toxins.
4. Some lymphocytes remain afterwards as **memory cells** (see 5.10).

Antibody binds to a specific antigen



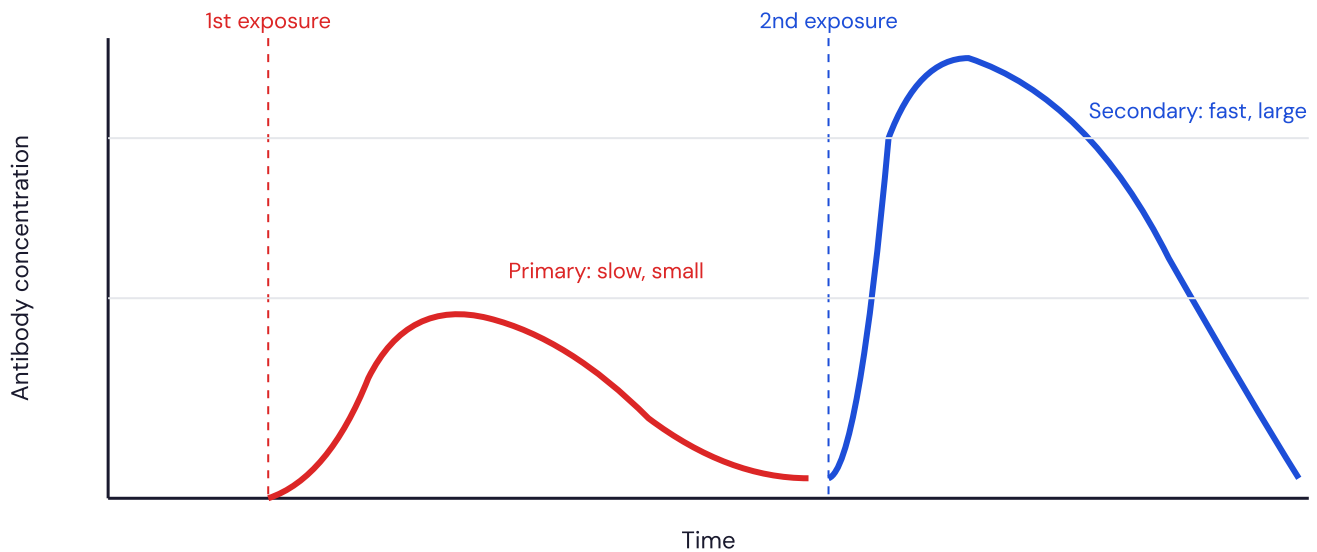
Exam Tip: Antibodies are **specific** — each antibody only fits one antigen. This is why the immune system must produce a different antibody for every pathogen, and why vaccines are disease-specific.

5.10 The Immune Response — Primary vs Secondary

The body responds differently the first time it meets a pathogen (**primary response**) compared with later exposures (**secondary response**). The secondary response is the basis of long-term immunity and of how vaccines work.

Feature	Primary response (1st exposure)	Secondary response (subsequent)
Speed	Slow — takes days to start (lymphocytes must find the antigen, match and divide).	Fast — memory cells multiply immediately.
Size	Small — few matching lymphocytes at first.	Large — many more antibodies produced.
Duration of symptoms	Person feels unwell while response builds.	Pathogen destroyed before symptoms develop.
Memory cells?	Produced for the first time.	Already present — rapidly divide.

Antibody concentration in primary and secondary responses



Exam Tip: In an exam, label both axes (time, antibody concentration), mark the two exposure arrows, and show the secondary curve as **higher, steeper** and starting **sooner** than the primary. Forgetting any of these loses marks.

5.11 Vaccines and Herd Immunity

A **vaccine** contains dead or weakened (inactivated) forms of a pathogen, or its antigens alone, introduced to stimulate an immune response *without causing the disease*.

How a vaccine works

1. Weakened/dead pathogen (or its antigens) is injected into the body.
2. The antigens trigger a **primary immune response**: lymphocytes produce antibodies and, crucially, **memory cells**.
3. If the live pathogen later enters the body, memory cells recognise the antigen immediately.
4. A fast, large **secondary response** destroys the pathogen before the person becomes ill. The person is *immune*.

Herd immunity

When a high proportion of a population is vaccinated, the pathogen cannot find enough susceptible hosts to spread. People who cannot be vaccinated (newborns, immune-compromised people, the very elderly) are protected indirectly — this is **herd immunity**. If vaccination rates fall below the threshold (around 90–95% for measles), outbreaks return.

Case study: Smallpox is the only human disease to be eradicated — a global vaccination programme ended natural transmission in 1977. It shows what vaccines are capable of. Measles, polio and rubella all fell dramatically after mass vaccination but have rebounded in areas where uptake has dropped.

5.12 Antibiotics and Resistance

Antibiotics are chemicals that kill bacteria or stop them reproducing, without harming human cells (e.g. penicillin, amoxicillin, tetracycline). They do **not work against viruses**.

Antibiotics target features unique to bacteria — for example, penicillin blocks the building of bacterial cell walls, which human cells don't have. Viruses have no cell wall and reproduce inside our own cells, so antibiotics cannot reach them. *Colds, flu and COVID are viral — antibiotics do nothing.*

Antibiotic resistance

Overuse and misuse of antibiotics has led to the evolution of **antibiotic-resistant bacteria** — a serious public health crisis.

1. A random **mutation** arises in a bacterial gene that happens to give some resistance to an antibiotic.
2. When the antibiotic is given, susceptible bacteria die but resistant ones survive — this is **natural selection**.
3. The resistant bacteria reproduce (every 20 minutes in some species) and pass the resistance gene to their descendants. They can also share it with other bacteria via plasmids.
4. Over time, an increasing proportion of the population is resistant; the antibiotic becomes useless.

Case study: MRSA (methicillin-resistant *Staphylococcus aureus*) has evolved resistance to most common antibiotics and causes hard-to-treat infections in hospitals. New "last-resort" antibiotics such as vancomycin are being lost to resistance too.

Slowing the evolution of resistance

- Doctors should only prescribe antibiotics when needed (not for viral infections).
- Patients must **complete the full course** — stopping early leaves the more resistant bacteria alive.
- Rotate antibiotics; use narrow-spectrum where possible.
- Control use of antibiotics in farming (where they are fed to livestock to promote growth).
- Improve hygiene to reduce the number of infections in the first place.
- Research new antibiotics.

5.13 Developing New Drugs

A new medicine goes through a long, staged development process to make sure it works and is safe. This typically takes 10–15 years and costs hundreds of millions of pounds.

Stage	What happens	Purpose
1. Preclinical — cells & tissues	Drug tested on cells and tissue cultures in the lab.	Does it have the intended effect? Is it cytotoxic?
2. Preclinical — animals	Tested on live animals (e.g. rats).	Check for toxicity and see how it is processed in a whole body.
3. Clinical trial — healthy volunteers	Low doses given to healthy humans, increased gradually.	Check for safety and side effects in humans; determine safe dose.
4. Clinical trial — patients	Given to people with the disease; double-blind, placebo-controlled , random allocation.	Is the drug actually effective at treating the disease? What is the optimum dose?

5. Peer review & publication	Results examined by independent scientists before publication.	Checks methods, statistics and conclusions. Prevents flawed or false claims.
6. Regulatory approval	e.g. MHRA (UK), FDA (USA) — agency licenses the drug.	Independent regulator decides if benefits outweigh risks.
7. Post-marketing surveillance	Ongoing monitoring after release.	Catches rare side effects only visible in very large populations.

Placebos and double-blind trials

- A **placebo** is a dummy treatment (e.g. a sugar pill) that looks identical to the real drug but has no active ingredient. It controls for the *placebo effect* — people can feel better just because they believe they've been treated.
- In a **single-blind** trial, patients don't know whether they got the drug or placebo.
- In a **double-blind** trial, neither patients nor the doctors assessing them know, so expectations cannot bias the measurements. The trial code is only broken after the data is collected.

Exam Tip: Spec favourite: "Why must new drugs be tested on healthy volunteers before patients?" → to check for *safety and side effects in humans* before testing *effectiveness* in people with the disease.

5.14 Monoclonal Antibodies (Higher Tier)

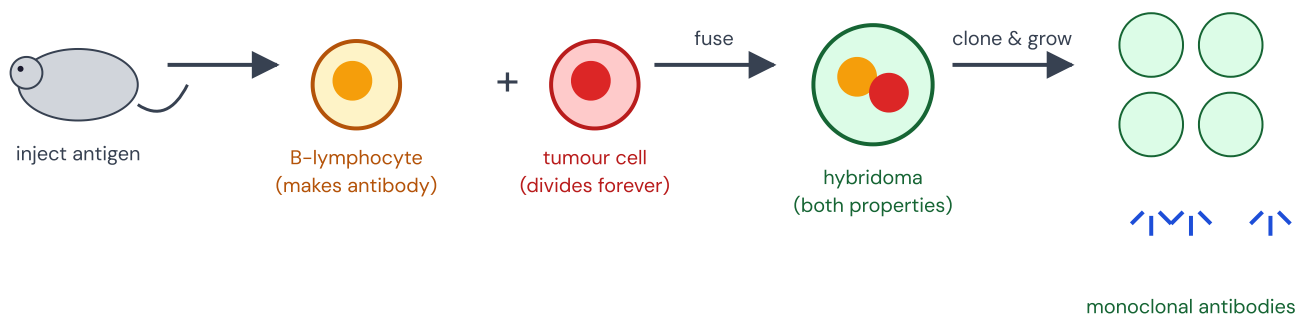
A **monoclonal antibody** is a pure population of identical antibodies, all specific to one antigen. They are made from a single clone of B-lymphocytes.

How they are made — hybridoma cells

B-lymphocytes make antibodies but don't divide in culture. Tumour cells divide indefinitely but don't make antibodies. By **fusing** the two you get a **hybridoma** cell that does both — making unlimited identical antibodies.

1. Inject a mouse with the target antigen — its immune system makes B-lymphocytes that produce the wanted antibody.
2. Extract B-lymphocytes from the mouse's spleen.
3. Fuse them with tumour cells (myelomas) to form hybridoma cells.
4. Screen hybridomas to find the one making the right antibody; clone it.
5. Grow the clone in large cultures — they produce unlimited quantities of a single (monoclonal) antibody.

Making monoclonal antibodies: B-lymphocyte + tumour cell → hybridoma



Uses of monoclonal antibodies

- **Pregnancy tests** — detect the hormone **hCG** (human chorionic gonadotropin) in urine. Antibodies in the test strip bind hCG and change colour.
- **Diagnosis** — e.g. detecting antigens of particular pathogens or cancer markers in blood.
- **Targeted drug delivery for cancer** — a toxin or radioactive substance is attached to an antibody that binds only to tumour cells, so the drug kills cancer cells while sparing healthy cells.
- **Research** — locating specific molecules inside cells.

Exam Tip: Don't confuse monoclonal antibodies (pure, all identical) with normal antibodies produced in an immune response (polyclonal — from many lymphocyte clones, directed at

many different antigens).

5.15 Non-Communicable Diseases (NCDs)

NCDs are not caused by pathogens and cannot be spread from person to person. They tend to develop slowly and are strongly influenced by **lifestyle and genetic** risk factors.

Cardiovascular disease (CVD)

Coronary heart disease (CHD) is the commonest form of CVD. The coronary arteries supply the heart muscle with oxygenated blood. Fatty deposits called **atheromas (plaques)** build up inside these arteries, narrowing them and reducing blood flow. If blood flow is blocked completely (e.g. by a clot forming on a plaque), part of the heart muscle dies — a **heart attack**.

Risk factors: smoking, high saturated-fat diet, high cholesterol, high blood pressure, obesity, lack of exercise, heavy alcohol use, stress, age, genetics, type 2 diabetes.

Cancer

Cancer is a disease caused by uncontrolled **cell division**. Cells grow into a lump called a **tumour**.

Type of tumour	Behaviour
Benign	Cells stay in one place inside a capsule; do not invade other tissue; rarely life-threatening (though can press on organs).
Malignant (cancerous)	Cells invade surrounding tissue and can break off, enter the blood and form new (secondary) tumours elsewhere — metastasis .

Risk factors: smoking (lung, throat, bladder cancers), UV radiation (skin cancer), ionising radiation, asbestos and other carcinogens, certain viruses (HPV → cervical cancer; HBV → liver cancer), obesity, alcohol, and inherited mutations (e.g. BRCA1/2 increases breast cancer risk).

Obesity

Obesity is excess body fat, measured by:

$$\text{Body Mass Index (BMI)} = \text{mass (kg)} \div \text{height}^2 (\text{m}^2)$$

BMI (kg/m ²)	Category
< 18.5	Underweight
18.5–24.9	Healthy weight
25.0–29.9	Overweight
30.0–39.9	Obese
≥ 40	Severely obese

BMI is a useful screening tool but has limits: very muscular people can have high BMI without being fat. A better measure of *where* fat is stored is the **waist-to-hip ratio** (waist circumference ÷ hip circumference). "Apple-shaped" fat around the belly (high ratio) is more dangerous to the heart than "pear-shaped" fat on hips/thighs.

Effects of alcohol

- **Liver:** long-term drinking causes fatty liver → alcoholic hepatitis (inflammation) → **cirrhosis** (scarring). A scarred liver cannot detoxify, make proteins or store glycogen; eventually fatal.
- **Brain:** impaired judgement and reaction times in the short term; long-term use damages brain cells, causing memory loss, mood problems and dementia-like conditions.
- Drinking during pregnancy → fetal alcohol syndrome (developmental/facial abnormalities).

Effects of smoking

- **Lungs:** tar damages cilia so mucus (and pathogens) build up → smoker's cough, bronchitis; tar contains carcinogens → **lung cancer**; breakdown of alveoli walls → **emphysema** (shortness of breath).

- **Circulation:** nicotine raises heart rate and blood pressure; carbon monoxide binds strongly to haemoglobin, reducing oxygen carriage; chemicals damage artery linings → faster build-up of atheromas → heart attack and stroke risk.
- In pregnancy: low birth weight, increased miscarriage/stillbirth.

5.16 Treating Cardiovascular Disease

Treatment	How it works	Pros	Cons
Lifestyle change	Stop smoking, eat less saturated fat, exercise, lose weight, moderate alcohol.	Free; addresses the cause; benefits whole body.	Requires long-term motivation; slow to take effect.
Statins	Drugs that reduce cholesterol made by the liver — slowing atheroma build-up.	Proven to reduce heart attacks/strokes long term; taken as a daily pill.	Side effects in some people (muscle pain, liver issues); must be taken for life.
Stents	A wire mesh tube pushed into a narrowed artery to hold it open.	Quick surgical fix; immediate improvement in blood flow.	Risk of blood clots, infection; does not treat underlying cause, which can re-block.
Coronary bypass	A healthy vessel (often from the leg) is grafted in to bypass a blocked coronary artery.	Effective for severe multi-vessel disease.	Major surgery, long recovery, risks of anaesthetic/infection.
Heart transplant	Diseased heart replaced with a donor heart.	Last resort for end-stage heart failure.	Shortage of donors; lifelong immunosuppressant drugs; rejection risk.

Exam Tip: Compare-and-contrast questions on heart treatments expect at least one biological benefit and one drawback per option. "Lifestyle is cheaper but harder to sustain; statins treat chemistry but have side effects; stents are fast but don't address cause" is a strong framing.

5.17 Risk Factors, Correlation and Causation

A **risk factor** is anything that increases the probability of developing a disease. Risk factors can be **modifiable** (diet, smoking, exercise) or **non-modifiable** (age, sex, genetics).

Reading data on risk

Epidemiologists study huge populations to pick out patterns. They express risk in several ways:

- **Incidence** — number of new cases in a year per 100 000 people.
- **Prevalence** — number of people currently with the disease.
- **Relative risk** — how much more likely you are to develop the disease if exposed (e.g. "smokers are 20× more likely to get lung cancer than non-smokers").

Correlation vs causation

A **correlation** is a statistical link between two variables. **Causation** is when one variable actually causes the other. *Correlation does not prove causation.*

- Ice-cream sales and drownings are correlated — but neither causes the other. The real cause is hot weather.
- Smoking and lung cancer were correlated for decades before causation was finally proven by controlled biological studies showing tar causes DNA damage in airway cells.

To go from correlation to causation, scientists look for:

1. A plausible biological mechanism (how could A cause B?).
2. A dose-response relationship (more A = more B).

3. Consistency across multiple large studies.
4. Experimental evidence — e.g. animal studies or natural experiments (e.g. smoking rates falling, lung cancer rates falling a decade later).

Case study: Many NCDs are strongly linked to *preventable* lifestyle factors. This is why public-health campaigns target smoking, diet, alcohol and exercise — modest changes across a population can prevent huge numbers of heart attacks, strokes and cancers.

Key Definitions — Glossary

Term	Definition
Health (WHO)	State of complete physical, mental and social wellbeing, not merely absence of disease.
Communicable disease	Disease caused by a pathogen and able to spread between organisms.
Non-communicable disease	Disease not caused by a pathogen and not spread between people (e.g. CVD, cancer, diabetes).
Pathogen	A microorganism that causes disease — bacteria, virus, fungus or protist.
Antigen	A chemical marker on the surface of a cell/pathogen recognised as foreign.
Antibody	A Y-shaped protein made by lymphocytes that binds specifically to one antigen.
Phagocyte	White blood cell that engulfs and digests pathogens (phagocytosis).
Lymphocyte	White blood cell that produces specific antibodies against a pathogen.

Memory cell	Lymphocyte formed after first exposure; gives long-term immunity via a fast secondary response.
Vaccine	Preparation of dead/weakened pathogen (or antigen) given to trigger a primary immune response and memory cells.
Herd immunity	Protection of the unvaccinated arising because so many others are immune that the pathogen cannot spread.
Antibiotic	Drug that kills bacteria or stops them reproducing; does not work on viruses.
Antibiotic resistance	Ability of bacteria to survive antibiotics; arises by mutation and natural selection.
Placebo	A dummy treatment used as a control in drug trials.
Double-blind trial	Clinical trial in which neither patient nor doctor knows who got the real drug vs placebo.
Monoclonal antibody	Identical antibodies from a single clone of lymphocytes, produced by hybridoma cells.
Atheroma (plaque)	Fatty deposit in the wall of an artery that narrows the lumen and can cause heart attack/stroke.
Benign tumour	Abnormal growth that does not invade other tissues or spread.
Malignant tumour	Cancerous growth that invades surrounding tissue and can metastasise.
BMI	Body mass index = $\text{mass (kg)} \div \text{height}^2 (\text{m}^2)$; screening tool for obesity.
Correlation	Statistical relationship between two variables; does not by itself prove one causes the other.